Weather-Aware Taxi Demand Prediction and Forecasting in New York City

Final Project Report

Submitted in partial fulfilment of the completion of
the course 23CSE452 BUSINESS ANALYTICS

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Declaration

We, the members of Team 12, declare that this report, "Weather-Aware Taxi Demand Prediction and Forecasting in New York City", describes work carried out by us for the course 23CSE452 Business Analytics under the guidance of Ms. Rohini S. The data was collected by the team from public sources, and the analysis, code, dashboards and results were produced by the team. All datasets, studies, algorithms, and software used are cited. This work has not been submitted elsewhere. Generative AI assistance is declared in Section 8.3.

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Abstract

The situation for taxi operators in New York City consists of losing money on waiting in empty areas for passengers in crowded areas. This project creates an analytics pipeline which predicts the locations and times of taxi pickups within the framework of 138 zones in Manhattan and Queens. The dataset used in the analysis was created independently from the requests to the trip data of the official NYC Taxi and Limousine Commission trip files and the Open-Meteo weather API: 23.3 million yellow and green taxi trips from September 2025 to February 2026 with the weather in the corresponding zone and hour. After preprocessing, the trips were transformed into a zone-hour dataset with 576,288 rows with calendar, weather and previous demand information. Exploration found out that there are 16 zones responsible for 50% of all the trips, the hour with the highest demand is Thursday, 6 pm, and heavy snow decreases pickup numbers by up to 75%. As a result of next-hour prediction with HistGradientBoosting model, a test MAE of 5.78 pickups per zone-hour and R^2 of 0.963, which is 37.9% error lower compared to the same-hour-last-week baseline, were achieved. Seasonal and intermittent demand forecast system decreased the day-ahead zone error by 16.0%, and association rule mining provided 152 holdout tested rules based on time, location and weather which detect the periods of high or low demand.

Keywords: taxi demand, NYC TLC, weather, gradient boosting, time-series forecasting, association rules, dashboard

Executive Summary

Problem: Demand for taxi services in New York is heterogeneous by zone, time, and weather. Fleets that are unable to anticipate changes will have cabs concentrated in busy areas while, at other places, cars will be cruising empty. The project displays where and when riders will be requesting a cab to fleet operators and dispatchers.

What we did: We gathered six months of trip records (23.3 million rides) and hourly weather for each pickup zone in Manhattan and Queens, we cleaned and studied the same data, and we built three tools: next-hour prediction modeling, day-ahead and week-ahead forecasting, and simple demand rules of the type "if this, then that". All of them are accessible via the interactive dashboard.

Key Findings: Demand is concentrated; 16 of 138 zones account for half of all pickups and Manhattan for 90.6%. The timings tend to repeat every week: Thursday, 6 pm is the busiest hour. Snow has a bigger impact than rain; during serious snowfall, pickups may be reduced by as much as 75%, while serious rain adds about 14%. A change in temperature has less than a 3% effect on demand.

The performance of KPIs is analyzed. The subsequent hour's prediction error has decreased from 9.31 to 5.78 (MAE) picks for each zone hour (37.9% improvement comparing with the reference level; R^2 0.963). The error in daily predictions has decreased from 9.31 to 7.82 (16.0% improvement) with the total figure for whole city relating to hourly demand now better by 16.5%. 152 demand regulations have passed the verification test on previously unseen weeks.

Expected value: Under assumptions stated in Section 6.3, operators with 10% market share could generate about USD 8,000 – 71,000 net value per month (USD 34,000 in the base case) under the stated assumptions. It is a rough estimate, not measured savings.

Recommendations: Schedule drivers' shifts and maintenance based on the week-ahead forecast; preposition cabs in commuter zones prior to 7 AM weekdays and nightlife zones Friday and Saturday evenings; position cabs in front of LaGuardia prior to 2 PM and JFK prior to 4 PM; limit active cab capacity during heavy snow warnings; and experiment with the dashboard using one dispatch team while monitoring forecast error weekly.

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Abbreviations

Abbreviation	**Meaning**
ARM	Association Rule Mining
DHR	Dynamic Harmonic Regression
DST	Daylight Saving Time
EDA	Exploratory Data Analysis
ETS	Exponential smoothing (Error, Trend, Seasonal)
HistGB	HistGradientBoosting Regressor
KPI	Key Performance Indicator
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MSTL	Multiple Seasonal-Trend decomposition (Loess)
RMSE	Root Mean Squared Error
R^2	Coefficient of determination
TLC	NYC Taxi and Limousine Commission
TSB	Teunter–Syntetos–Babai method
WAPE	Weighted Absolute Percentage Error

1. Introduction, Business Objectives and Literature Review

1.1 Context and scope

The NYC Taxi and Limousine Commission (TLC) collects all data for trips of green and yellow taxis in the City of New York and provides the data as open data. Unfortunately, supply and demand matching is difficult, especially since demand for taxis increases and decreases by location, time, and even the weather. As with many business problems, demand fluctuation results in a shortage of cabs in some areas and too many idle cabs in others. In response to this, our project has created three models of demand prediction that guide cab allocation and shift scheduling, and weather-based adjustments of cab allocation. This project builds on existing demand prediction models using data from before 2020 on Manhattan's yellow taxis. Using a single city-wide weather prediction model, these demand prediction models rely on demand patterns from pre-pandemic, pre-congestion pricing data.

Our scope includes all the TLC 138 taxi zones, with half, 69 zones, being in Manhattan and the other half in Queens, for a period from September 1, 2025, to February 28, 2026. Our objective is to determine and predict demand on a zone by zone and hour by hour basis. All other ride demand factors such as ride hailing and fare information, and even unfulfilled taxi demand and trip drop off locations are outside our scope.

We are primarily concerned with predicting and fulfilling demand based upon our analysis and model. This is done based on five analytical questions:

Q1 - Where is demand concentrated?

Q2 - When does it change, and how much does rain, snow and temperature matter?

Q3 - How accurately can the next hour be predicted?

Q4 - How accurately can pickups be forecast 24 hours and 7 days ahead?

Q5 - Which conditions reliably go with high or low demand?

The KPIs and the diagram of how the unit pieces fit together are in Table 1 and Figure 1, respectively.

Business objective	KPI / unit	Baseline / target	Relevant unit or method
Find priority zones and demand drivers (Q1, Q2)	Share of trips in top zones (%); weather effect (% vs. normal dry hour)	None / identify zones covering 50% and 80% of trips; quantify each weather effect	U1 – EDA
Predict next-hour pickups for repositioning (Q3)	MAE, RMSE (pickups per zone-hour); R^2	Naive same-hour-last-week MAE 9.31, R^2 0.878 / ≥ 20% lower MAE, $R^2 \geq 0.90$	U2 – regression models
Forecast day- and week-ahead pickups for shifts (Q4)	Zone-hour MAE; city-hour MAE (pickups)	Weekly naive MAE 9.31 (zone), 777.7 (city) / ≥ 10% lower MAE	U3 M1 – time series
Find conditions that trigger high or low demand (Q5)	Rules with test confidence ≥ 0.50 and test lift ≥ 1.20	None / stable, non-redundant rules checked on unseen weeks	U3 M2 – association rules

Business objective	KPI / unit	Baseline / target	Relevant unit or method
Give planners one tool for all results	Working dashboard with filters and live tools	None / all unit outputs in one application	Application

Table 1. Business objectives, KPIs and success criteria

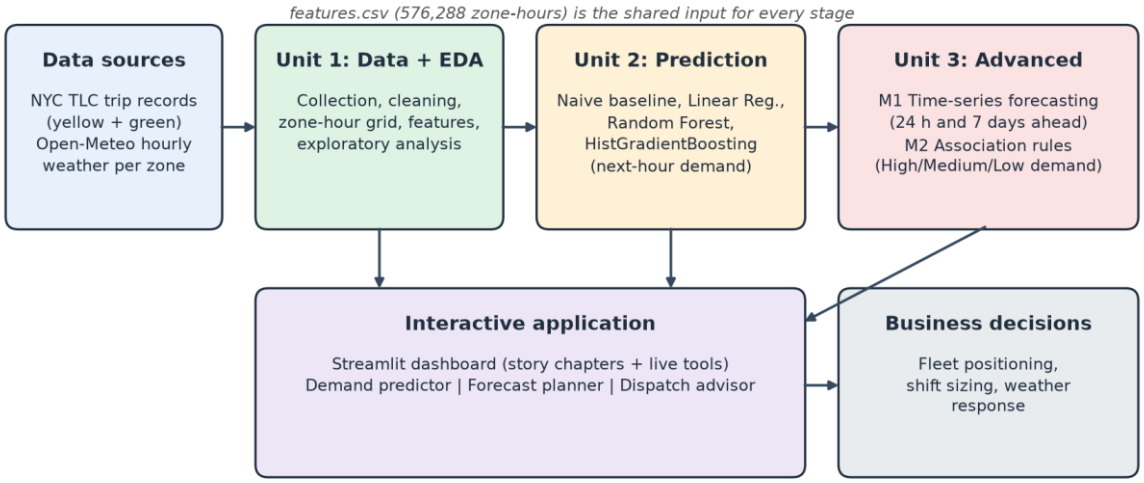


Figure 1. End-to-end workflow linking data collection, Units 1–3 and the interactive application.

1.2 Related analytics work and research gap

Correa and Moyano [1] predicted hourly yellow, green and Uber pickups by zip code in all five boroughs based on 90 million trips in 2014. XGBoost achieved the best results (RMSE 38.51, R^2 0.97), but the dataset is over ten years old and weather has not been considered. Rajendran [2] forecasted demand for a future air-taxi service based on NYC taxi trips with time and weather features. Gradient boosting outperformed logistic regression, neural networks, and Random Forest. Location, time of the day and weekday were the main predictors; the problem was solved through classification, not forecasting of hourly pickups. Studies demonstrated that rain influences both NYC taxi ridership and the balance between supply and demand, therefore the actual number of pickups is an imperfect proxy for demand. They used LSTM network for forecasting real-time NYC taxi demand and combined Poisson and ARIMA models for forecasting taxi-stand demand in Porto. Both studies show that recent demand and its daily/weekly seasonality carry almost all information.

Gaps Found: Recent data, green taxis and the borough of Queens have not been studied. Weather is usually obtained from one station. The incremental value of weather variables over the past demand has never been evaluated. Sparse areas with many zero hours are usually disregarded. Only a few works provide interpretable rules or a usable application of their model. Reasons why our approaches work: boosted trees are already used successfully. Poisson loss function is appropriate for counts; seasonal models are appropriate for the strong seasonality; TSB method is designed for intermittent data; association rules provide useful IF-THEN statements for dispatchers. A numerical comparison with these papers is provided in Section 6.2.

2. Dataset Collection, Validation and Preparation

2.1 Provenance, coverage and challenges

The data has been collected using the team's Python code (scripts/data_collection.ipynb), without relying on any existing repository dataset. Trips were extracted via programmatic HTTP requests to the Parquet files provided by the TLC monthly in yellow/green taxis [3]. These files contain publicly available open data. The zone lookup list and shapefile of TLC contained 138 zones, and the area weighted centroids have been calculated with pyproj. Apparent temperature, rain, snowfall and precipitation probability in hourly resolution for each centroid is extracted using the Open-Meteo Historical Forecast REST API [4] (open access, CC BY 4.0). The size and row count of each TLC file are recorded in taxi_collection_summary.csv for versioning purposes. The 3.2 GB collected file is stored in the private Kaggle dataset.

Coverage: Out of 24,562,021 trips in 12 monthly files, 23,319,012 started in either Manhattan or Queens. Removing the 33 trips outside of the relevant period resulted in 23,318,979 trips and 14 columns in the collected file. Cleaning resulted in 22,579,793 trips, which were then aggregated to form a complete grid of 138 zones x 4,344 hours (599,472 rows). Removing the 168 hours of the warm-up week, the final features table features.csv contains 576,288 rows with 17 variables in zone-hour resolution from 8 September 2025 to 28 February 2026. The target is trip_count. The different types of variables comprise integers (such as IDs, counts, calendar), floats (weather, rolling means), Booleans (weekend, holiday), and one datetime variable. Information regarding the sample can be found in Table 2, whereas Tables 4 and A contain data dictionary entries.

Table 2. Sample of the final dataset (selected columns)

Zone (ID)	pickup_hour	Trips	Temp (°C)	Rain (mm)	Snow (cm)	Holiday	Lag 1h	Lag 24h	Lag 168h
Midtown Center (161)	2026-02-12 18:00	593	-5.7	0.0	0.00	No	591	606	431
JFK Airport (132)	2025-10-16 16:00	434	9.5	0.0	0.00	No	366	464	330
East Village (79)	2025-12-13 01:00	845	-5.9	0.0	0.00	No	711	289	861
Astoria (7)	2026-01-20 08:00	40	-14.6	0.0	0.00	No	35	14	16
LaGuardia Airport (138)	2026-02-23 14:00	0	-7.7	0.1	0.21	No	0	128	197
Upper East Side N. (236)	2025-11-27 09:00	145	0.0	0.0	0.00	Yes	121	339	422

Issues faced during Data Collection:

1. The NYC Open Data API did not offer any information about the yellow cab data for 2025 or 2026, so we had to download the available monthly data files.
2. The Open-Meteo API restricts calls to 600 calls per minute; therefore, we formed 14 batch requests for the data from 10 zones, leaving a 15-second interval between the requests.
3. Rainfall and snow have been recorded for the last hour hence we shifted the data collection for one hour back.
4. The 3.2GB file was processed in units of one month, and only relevant columns were used.
5. The yellow and green versions of files named their columns differently, and therefore the columns were standardized.
6. The hour of 01:00 was retained while implementing Daylight Savings time on November 2nd of 2025; hence, we labeled that time and we do not use synthetic data.

2.2 Data quality, privacy and ethics

Table 3. Data quality checks, issues found and treatment

S. No.	Check	Finding	Treatment / retained
1	Exact duplicate trips	276,749 (1.2%)	Removed; first copy kept
2	Drop-off at or before pickup	329,654 (1.4%)	Removed (meter clock errors)
3	Trip longer than 6 hours	6,826 (0.03%)	Removed (unclosed meters)
4	Unknown / outside-NYC drop-off zone	125,957 (0.5%)	Removed; 22,579,793 trips kept (96.83%)
5	Missing values	None; weather present for all 599,472 zone-hours	No imputation needed
6	Zero-pickup zone-hours	27.7% (Manhattan 11.6%, Queens 43.7%)	Kept as true zeros
7	Repeated DST hour (2 Nov 2025, 01:00)	9,166 trips in one hour label	Kept; halved only in the time-series target
8	Imbalance and weather resolution	Manhattan 90.6% of trips; 40 zones < 1 pickup/hour; 50 weather cells for 138 zones	Kept; zero-aware methods and group-level error checks
9	Final-table validation	15 of 15 checks passed (schema, ranges, joins, totals)	576,288 rows confirmed

Privacy and compliance: The TLC trip records are public open data released by the City of New York under its Open Data Law. They contain no names, phone numbers, licence plates, medallion or driver IDs, payment details or exact coordinates; locations are given only as taxi zones. We kept only five trip fields and aggregated everything to counts per zone and hour. Because no information about an identifiable person is processed, the data falls outside both US privacy rules and India's Digital Personal Data Protection Act, 2023. The US has no single federal privacy law, and New York's SHIELD Act covers private information such as account or

identity numbers, none of which is present. We still followed the principles common to both purpose limitation and data minimisation, and made no attempt to re-identify any trip, driver or rider. The raw file is shared only with team members; large files are not uploaded to GitHub, and no API keys or secrets are stored.

Biases and limitations: pickups understate demand when cabs are scarce, for example during snowstorms; trips by ride-hailing companies are not included; only autumn and winter months are covered; and weather is taken at zone centroids and excludes short convective showers.

2.3 Transformations and usable dataset

The overall data cleaning was as follows: Checks 1-4 from Table 3 were applied (23,318,979 to 22,579,793); pickup times were rounded down to the nearest hour; totals were counted for each zone. The dataset was padded by combining the 599,472 zone-hour combinations found in the weather table with 0. The four weather data columns were merged. Additional columns for time and date information were created. Time and date information were also converted into several lags and rolling averages for each zone. To avoid including empty data points in the initial 168-hour lag analysis, the first week of data (23,184 rows) was dropped (599,472 to 576,288).

For this analysis, preprocessing of the data for each model was performed on the training data. This includes: one-hot encoding for the Linear Regression and Random Forest models; ordinal categorical encoding for HistGradientBoosting; demand and temperature binning; zone clustering; and the recovery of the data for the warm-up week. For training and testing the models, data were split into two periods. The first period was from 8 Sep 2025 to 25 Jan 2026 (460,920 rows) for the training data and the second period was from 25 Jan 2026 to 28 Feb 2026 (115,368 rows) for the testing data. Each model utilized the same chronological split. Data points in each zone were not split up or reordered in any way.

Table 4. Data dictionary and mapping of fields to units and methods

Variable	Type / meaning	Origin / preparation	Used in
PULocationID	Integer; pickup zone (138)	TLC trips, Manhattan/Queens filter	U1, U2, M1, M2 (via cluster)
pickup_hour	Datetime; start of hour (NYC time)	Pickup time rounded down	All (index, split)
trip_count	Integer; clean pickups (target)	Count of cleaned trips, zero-filled	U1, U2, M1 target, M2 demand class
apparent_temperature	Float; feels-like (°C)	Open-Meteo at zone centroid	U1, U2, M2 (3 bins)
rain / snowfall	Float; mm / cm in the hour	Open-Meteo, shifted to starting hour	U1, U2, M2 (yes/no)
precipitation_probability	Integer; %	Open-Meteo, shifted to starting hour	U1, U2
hour, day_of_week, month	Integers; calendar position	Derived from pickup_hour	U1, U2, M1 (seasonal slots), M2
is_weekend / is_holiday	Boolean flags	Derived; holidays package	U1, U2 (holiday), M2

Variable	Type / meaning	Origin / preparation	Used in
trip_count_lag_1h/24h/168h	Integers; earlier demand	Shift within sorted zone	U1, U2
trip_count_roll_3h/24h	Floats; mean of previous 3/24 h	Shifted rolling mean within zone	U1, U2

3. Exploratory Data Analysis and Visualization (Unit 1)

EDA answers Q1 and Q2 and pre-processes features for modelling. All figures are done in Python with Matplotlib and Seaborn based on features.csv (sources of data are specified in Section 7.2). The hours with zero values are kept throughout.

Distribution: Hourly pickups have extremely skewed distribution: mean is 37.7, median 4 and the maximum value is 1,122; 27.7% of zone-hours have zero pickups. In 135 out of 136 operational zones variance is larger than the mean (median ratio is 3.53). It suggests the count-based loss and that MAPE will be highly volatile (Appendix B, Figure 9).

Where demand is generated: Only 16 zones among 138 generate half of all the pickups and 33 zones generate 80% (Figure 2a). The Gini Coefficient is 0.71, 0.49 for Manhattan and 0.84 for Queens. On average, the zones in Manhattan have 68.3 hourly pickups whereas those in Queens have 7.1 hourly pickups. The two airports are responsible for 6.5% of all trips, with the largest being JFK airport at 4 pm with 380.9 trips per hour, and LaGuardia airport at 2 pm with 258.6 trips per hour. The weekly time series of 168 hours was clustered by k-means (k = 3, silhouette = 0.30), which gives rise to three types of activity zones (Figure 2b):

i) core business and airports represent 58 clusters, responsible for 84.3% of all trips, with the peak at daytime;

ii) the nightlife groups consist of 10 clusters, making up 10.7% of all trips, with the peak at night, done mostly on Fridays and Saturdays;

iii) residential urban areas consist of 30 clusters, accounting for 4.8% of trips, with a peak certainly observed in the morning from 7 to 9 am on weekdays.

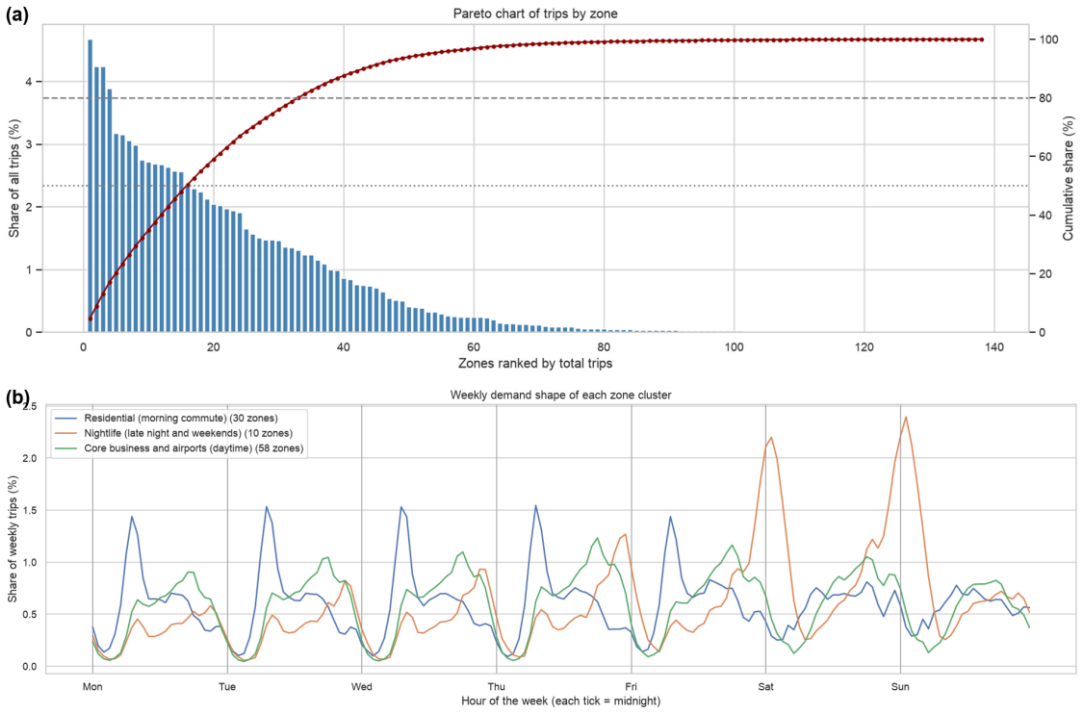


Figure 2. (a) Pareto chart: each zone's share of trips (bars, %) and cumulative share (line, %). (b) Average weekly shape of each zone cluster (share of weekly trips per hour, %). Tool: Python/Matplotlib.

Time-based demand analysis: The peak hour is Thursday at 6 pm (approx. 10,770 pickups per hour) and the lowest one is Tuesday at 3 am (approx. 280). Weeknight periods from midnight to 3 am are 4.7 times busier than weekday night periods (heatmap in Appendix B, Figure 10). Weekly cycles are highly regular, with a median correlation of 0.90 between weekly cycle of each zone and its expected weekly cycle. Daily trips have fallen by 15% between the first four weeks and the last four weeks. Out of 174 days, 18 days were considered abnormal, and all had an obvious reason (Appendix B, Figure 11): holidays periods (Christmas Day -57.6%, Thanksgiving -44.8%), snowstorms (23 February -78%) and pre-Christmas rush (+32%).

Weather: Since snow precipitates only in winter and rain is more probable during some hours, weather was measured as a demand ratio: real trips divided by the average value of dry hours with the same zone, hour, day type and month. Light rain increases demand somewhat (+5%) and very heavy rain (+14.1% above 5 mm/hour), and the effect occurs mainly in the morning (+15%-+19%). Snow decreases demand by 25%, 44% and 75% when becoming stronger, and snow precipitations decrease demand by airport about 60%. Temperature is within ±3% of expected (Figure 3).

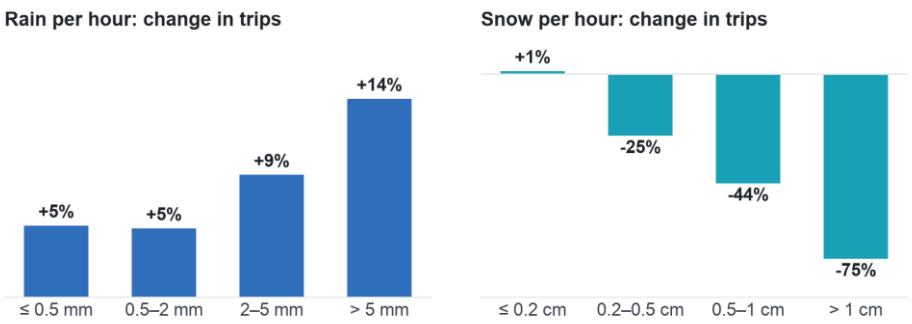


Figure 3. Demand ratio by rain intensity (left) and snowfall intensity (right); 1.0 is a normal dry hour, and labels show the % change and the number of zone-hours. Tool: Python/Matplotlib.

Relationships among features and implications for modelling: Lag features correlate 0.91-0.96 with the target and their recomputation gave zero mismatches, so there is no leakage. Zone alone accounts for 55% of the variance and zone plus hour 84%. The 1-hour lag and 3-hour mean overlap greatly (VIF of 28.4 and 19.8). Thus, the conclusions drawn were past demand, zone and hour as key predictors, weather checked separately; loss function by count and error reporting by zone group; time-splitting; seasonal prediction with special model for zones with low demand; and time band, day type, weather flags and zone cluster as the feature set for rule mining.

4. Feature Engineering and Predictive Modelling (Unit 2)

Target and predictors: The models predict trip_count in the next hour using 14 variables: zone; hour, weekday, month and holiday; four weather parameters; three lags; and two rolling means. Weekend is excluded since it duplicates weekday. Dimensionality reduction was not necessary for 14 variables. Models: baseline model that replicates zone and hour one week earlier; Linear Regression with one-hot encoding; Random Forest (150 trees, max_depth=20, min_samples_leaf=2, max_features=sqrt); and HistGradientBoosting (loss='poisson', learning_rate=0.08, n_iter=200, max_leaf_nodes=31, L2=1.0, categorical zones and date features). Random seed used is 42.

Protocol and Leakage Audit: The model was trained using the first 80% of timestamps and evaluated using the last 20% (chronological hold-out strategy). After conducting the tests on 80/20, 70/30 and 60/40 splits, the results achieved by 80/20 were better and hence the models were trained based on this split. The encoding and imputation steps are executed using a pipeline solely on the training set; only the lag and moving average are created using the past hours from the same location and at no point is the training set and testing set mixed. The one-hour lag requires the count from the previous hour; hence the model can only be applied for one-hour-ahead predictions while further forecasting is discussed in Section 5.1.

Table 5. Model comparison on the same held-out test period

Model	Validation scheme	Test metrics	Interpretability / bias evidence
Naive 168-hour baseline	Chronological 80/20 hold-out	MAE = 9.31, RMSE = 24.06, R^2 = 0.878	Fully transparent; misses recent changes
Linear Regression	Chronological 80/20 hold-out	MAE = 7.49, RMSE = 15.70, R^2 = 0.948, MAPE = 87.5%	Coefficients; lags highly collinear
Random Forest	Chronological 80/20 hold-out	MAE = 6.80, RMSE = 15.26, R^2 = 0.951, MAPE = 54.5%	Impurity importance; largest train–test gap (6.20 → 6.80)
HistGradientBoosting (selected)	80/20 hold-out; 70/30 and 60/40 checks	MAE = 5.78, RMSE = 13.16, R^2 = 0.963, MAPE = 44.8%	Permutation importance; errors by hour, borough and service area

HistGradientBoosting scores highest in all metrics, having 37.9% lower MAE and 45.3% lower RMSE when compared to the baseline (non-zero hours are only used for MAPE). All scores resemble each other, with the MAE of training being 5.54 and of testing being 5.78, leading to no overfitting. The choice of this model is based on its predicted accuracy, Poisson loss for count values, built-in ability to deal with categorical values, and its efficiency in handling larger datasets by segregating them into bins.

Issue of error analysis: the model under-predicts the highest values (refer to Appendix B, Figure 12). Over-predictions appear more often (62%) compared to under-predictions, but under-prediction values are bigger in magnitude compared to errors. The lowest values of errors are observed at 3 am (MAE 1.9), while the highest ones are seen in the evening, around 10 pm (MAE 9.0).

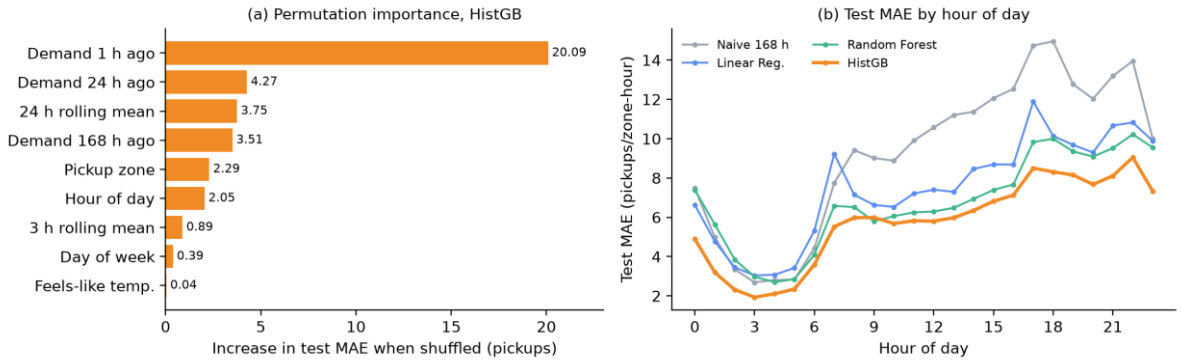


Figure 4. (a) Permutation importance for HistGradientBoosting: increase in test MAE when a feature is shuffled. (b) Test MAE by hour of day for all models. Tool: Python/Matplotlib.

Interpretation: By far, the highest reported MAE occurs when including demand from the previous hour, which results in an increase of 20.1 pickups. The next most important determinants are demand from the previous day, demand from 24 hours ago, demand from a week ago, and hour and zone. Weather does not contribute to anything on this horizon. Omitting the four weather variables leads to a lower MAE of 5.67 as the last hour reflects the weather. Weather variables are part of the regression equation but bear no causal relation. A small subgroup test shows that MAE is lower in Queens (mean of 6.3 pickups) than in Manhattan

(mean of 61.1 pickups) 9.52 and 2.05 respectively and equals 29.1 in the airports. The relative error of prediction is thus about twice as high in Queens (32%) than in Manhattan (16%).

5. Advanced Analytics (Unit 3): Two Selected Methods

5.1 Method 1 — Time-Series Analysis

Business problem: How many pickups will each zone get during each hour of next 24 hours as well as 7 days using only available data based on the moment of forecasting? This will help plan drivers' rosters and maintenance.

Inputs and preparation: All 138 zones for 4,344 hours (599,472 rows). The warm-up week that was skipped was found in the 168-hour main column. The 22,579,793 trips were reconciled with the results obtained and the target trip_count_adjusted obtained by merging the DST hour was calculated. Six aggregate series were created (all zones, Manhattan, Queens, Yellow Zone, Boro Zone, Airports) as well as daily ones. According to training profiles, there are 83 regular zones, 53 intermittent zones, and 2 zones without demand predicted.

Justification of the method: the daily periodicity is 0.94 and weekly periodicity is 0.76, while the autocorrelation values are 0.88 for 24 hours and 0.89 for 168 hours. The ADF and KPSS tests found that after applying logarithms and differencing the seasonal series with a length of 168 hours the stationarity condition met, so the weekly variation becomes stable. Algorithm and variables: four methods for each series were investigated, which include:

1. Adaptive seasonality, where a weekly pattern from four weeks ago B(h) is multiplied by attenuation F(h) = B(h)·[1 + (r − 1)·e^(−h/48)], where r is in the range from 0.5 to 1.5.
2. Mean value of the same weekday and hour of the previous four weeks.
3. TSB algorithm for intermittent zones: F = p·z, where the probability of pick-up p is adjusted every hour (β = 0.05), and the positive size of pick-ups z is adjusted upon occurrence of pick-ups (α = 0.10).
4. Naive weekly.

Four 168-hour rolling-origin folds prior to the test period selected for the approach with the minimum MAE per series. There are 88 adaptive, 25 four-week mean, 17 TSB, and 8 naive zones. The zone predictions are summed to ensure consistency of the total; then 80% and 95% prediction intervals are calibrated in two further folds. The source code can be found in models/time_series/model_1/ (run_time_series.py, with tests in test_time_series.py).

Table 6. Forecast accuracy on the test period (MAE, pickups per hour)

Evaluation level	Weekly naive	Model 1	Error reduction
Each zone, each hour (24 h blocks / 168 h blocks)	9.31	7.82 / 8.03	16.0% / 13.7%
City total, direct forecast / summed zone forecasts	777.71	678.21 / 649.37	12.8% / 16.5%

Evaluation level	Weekly naive	Model 1	Error reduction
Manhattan / Queens / Airports totals (bottom-up)	713.8 / 118.1 / 89.6	591.7 / 91.1 / 71.2	17.1% / 22.8% / 20.5%
City total per day (daily model)	14,270	12,774	10.5%

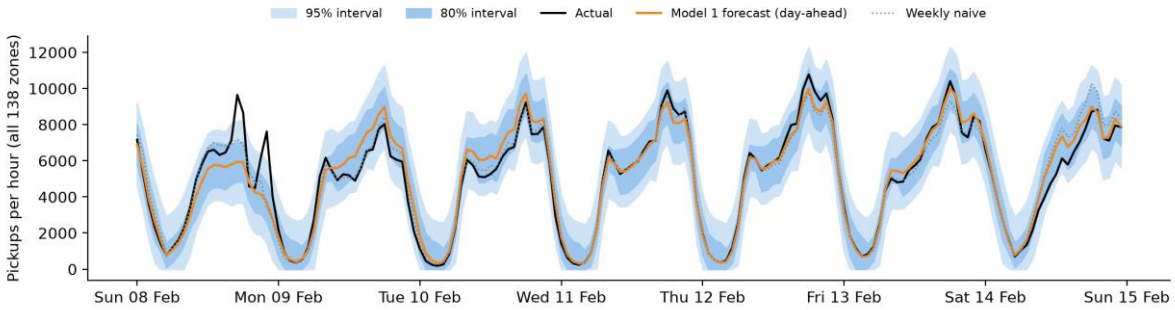


Figure 5. Actual city-wide hourly pickups against the Model 1 day-ahead forecast, with 80% and 95% intervals and the weekly naive forecast, test week 8–14 February 2026. Tool: Python/Matplotlib.

Results: The primary forecasting method was able to follow daily consumption peaks and valleys and did better than the baseline naive approach over all levels of observation. The highest level of improvement was seen in Queens and Boro Zone, where level adaptation and TSB can represent sporadic consumption. City's R^2 was equal to 0.862 and the obtained prediction intervals were able to cover 83.3% of the data in day ahead forecast and 79.7% in week ahead forecast. Only the prediction intervals for Queens turned out to be narrower than expected, allowing for 73.4%. Two additional forecasting models per zone were implemented using a different chronological split - Dynamic Harmonic Regression (DHR) and MSTL with Exponential Smoothing (ETS) methods. DHR showed the test results with MAE of 11.98 and RMSE of 28.55 and R^2 of 0.8314. MSTL + ETS has shown MAE of 11.35 and RMSE of 29.51 with R^2 of 0.8198. DHR has a higher MAE and lower RMSE compared to MSTL + ETS. Based on R^2, DHR performed better and their test results for WAPE were equal to 30.51% and 33.24% respectively. DHR had lower WAPE due to this showing better weighted accuracy.

Error analysis: Both of the extra models had issues in handling the strange reduction of demand around February 22-23, 2026. It was found that there was a massive NYC blizzard occurring during this time frame. An emergency travel ban prevented all non-essential vehicles from driving on the city roads, highways, bridges, and tunnels between 9 PM on February 22 to noon on February 23 [5]. Actual demand for taxis drastically fell, but the models still forecasted using the recurring pattern in the past, hence producing large errors.

Limitations: Only six months of data was not enough to detect annual seasonality. It was impossible for the forecasting models to predict any unforeseen occurrences such as travel bans or disruptions unless there is an occurrence of input included in the model.

5.2 Method 2 - Association Rule Mining

The business question: What are the relationships between time, calendar, weather, and zone type that predict high, moderate, or low levels of demand?

Data preparation: Each transaction is one zone hour per family per item and involves the following variables: time period (late night, early morning, AM rush, midday, PM rush, evening); weekend/holiday; federal holiday/normal day; rain/no rain; snow/no snow; temperature tercile; and zone cluster. The three demand classes are Low, Medium or High, cut at each zone's own training-quantiles 1/3 and 2/3. Of the 96 active zones (99.66% of pickups), the only one excluded was that of the adjustment hour for daylight-saving time, therefore creating 320,544 training and 80,256 test transactions. Bins and clusters (57 daytime, 29 morning-commute, 10 nightlife) were fitted on training data only. Why this method: rules capture combinations and are easy for dispatchers to read.

Algorithms and parameters: Apriori [6] and FP-Growth were run with minimum support of 0.05, 0.01, and 0.005. Only rules with single demand items in the consequent, confidence ≥ 0.50, and lift ≥ 1.20 were considered. Both algorithms discovered the same 6,719 itemsets and 897 rules at minimum support of 0.005, however FP-Growth ran in 1.3 seconds versus 28.5 of Apriori (see Figure 6a), so FP-Growth was chosen. Validation: every rule was tested on the test weeks and only those were retained for which the rule's conditions occurred at least 100 times with confidence and lift ≥ 0.50, and ≥ 1.20 respectively. This resulted in 632 rules. Next, all rules that increased conditions without increasing confidence were pruned, resulting in 152 rules (92 Low, 49 High, 11 Medium; see Figure 6b).

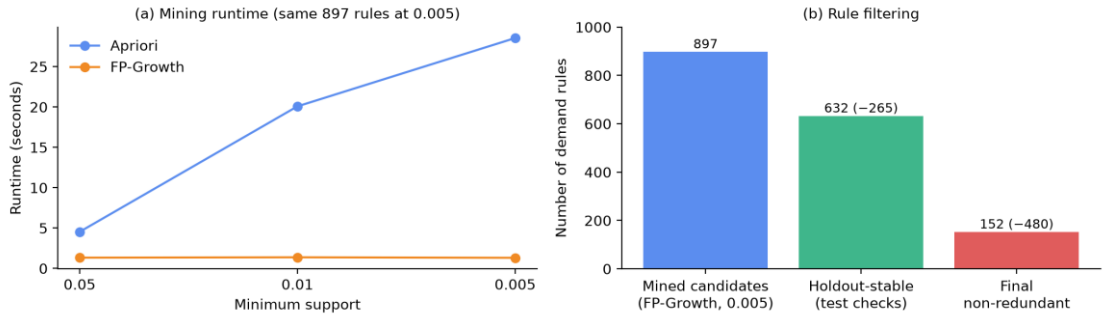


Figure 6. (a) Runtime of Apriori and FP-Growth by minimum support. (b) Rule filtering from 897 candidates to 632 holdout-stable and 152 non-redundant rules. Tool: Python/Matplotlib.

Table 7. Strongest validated rules for each demand class (ranked by test lift)

IF (conditions)	THEN	Train conf.	Test conf.	Test lift	Test count
weekday, regular day, AM rush, commuter zone	High	0.65	0.73	2.70	2,088
regular day, evening, no rain, nightlife zone	High	0.64	0.63	2.32	1,258
weekend, no snow, nightlife zone	High	0.51	0.55	2.03	2,012
weekday, regular day, late night, daytime zone	Low	0.92	0.95	2.27	6,840

IF (conditions)	THEN	Train conf.	Test conf.	Test lift	Test count
weekday, no rain, midday, daytime zone	Medium	0.51	0.61	1.95	7,864

Interpretation: Weekday mornings for commuters are High 73% of the time (2.7x more often than usual). Evening for nightlife zones is High on weeknights (63%) and weekends (55%). Demand on late nights on weekdays in business zones is almost always Low (95%), offering opportunities for breaks and maintenance. The test confidence was usually higher than the training confidence, and thus the rules applied even to the unseen weeks. 49.7% of the pruned conditions were weather-related, while time bands were never pruned.

Limitations: the classes are relative to each zone, not number of cabs; the rules demonstrate only correlation; the rules for warm weather cannot be tested; and 42 zones with few cabs were excluded.

6. Comparative Analysis and Business Impact

6.1 Comparison of the two Unit 3 methods

Criterion	Method 1: Time-Series Analysis	Method 2: Association Rule Mining
Question / output	How many pickups in the next 24 h / 7 days; forecasts with intervals	When demand is High/Medium/Low for a zone; 152 IF-THEN rules
Inputs	Past hourly counts of 138 zones and 6 aggregates	400,800 zone-hour transactions with 8 categorical items
Validation	Rolling-origin selection; MAE, WAPE and interval coverage on test weeks	Test-week count, confidence and lift; redundancy pruning
Strengths / limits	Gives quantities for staffing; blind to storms and events	Easy to act on; relative classes, association only

Table 8. Comparison of the two Unit 3 methods

The techniques used in the study do not lend themselves to being ranked. The forecasting method estimates the number of cabs required in each zone, while the rules determine the time when an area will be considered busy. Both time and place are critical according to Units 1, 2, and 3 consensuses. Zone and hour account for 84% of the variance in EDA, lag, zone, and hour rank high in Unit 2. Weekly and daily cycles account for the predictions, while time band represents an important element of every rule. Morning-commute and nightlife peaks could be seen both as clusters of EDA and among the strongest rules. The situation is paradoxical as snow reduces the number of pickups by about 75%, but weather impacted the next-hour model very little and is often excluded from rules. Snowy hours are very rare (approximately 2.5%) and last-hour count captures them already; while snow can lead to failure during day-ahead forecasts.

6.2 Comparison with existing studies

Table 9. Comparison with existing studies

Criterion	Correa and Moyano [1]	Rajendran [2]	Current project
Data and setting	90 M Yellow, Green and Uber trips; five boroughs; 2014	NYC taxi and air-taxi demand with time and weather factors	23.3 M Yellow and Green trips; Manhattan and Queens; 2025–26

Criterion	Correa and Moyano [1]	Rajendran [2]	Current project
Prediction approach	Hourly demand prediction	Classification of air-taxi demand	Zone-level hourly demand regression and time-series forecasting
Machine learning methods	Regression, Random Forest, XGBoost	Logistic Regression, Neural Network, Random Forest, Gradient Boosting	Linear Regression, Random Forest, HistGradientBoosting
Best reported ML result	XGBoost best; RMSE 38.51, R^2 0.97	Gradient boosting best; location, time, weekday key	HistGB best; RMSE 13.16, R^2 0.963; lags, zone, hour key
Time-series methods	Not reported in the cited study	Not reported in the cited study	Dynamic Harmonic Regression (DHR), MSTL + ETS, seasonal/TSB forecasting
Temporal modelling	Hourly demand patterns	Time and weekday factors	Chronological validation, daily and weekly seasonality, lag features and rolling-origin evaluation
Additional forecasting evaluation	—	—	DHR: test MAE 11.98, WAPE 30.51%; MSTL + ETS: test MAE 11.35, WAPE 33.24%

The findings of these studies show that boosted trees perform better in their use and that the two most important factors are location and period. However, due to using different methods for prediction by the authors, the used numbers cannot be compared with each other directly. Correa and Moyano predicted a bigger area of the towns using their dataset from 2014; Rajendran used a more complicated classification error. Therefore, our project is different by the addition of the following: we used up-to-date data after the implementation of congestion pricing policy about green taxi cabs activity in New York City in Queens area, basic weather data that can be important in terms of taxi activity, several forecasting techniques used for low-density areas and rules that have been used for the purpose of the writing the paper.

6.3 KPI-based business impact and scenarios

Achieved measures associated with Table 1: 16 zones cover 50% of trips (U1); the next-hour error is 37.9% smaller than the baseline (U2); day-ahead error by zones and city is 16.0% and 16.5% smaller (U3 M1); and 152 rules were valid during test weeks (U3 M2).

Hypothetical value: the improvement in day-ahead prediction by 1.49 pickups per zone-hour (9.31 - 7.82) implies 205 pickups per hour, or 4,930 per day, being predicted properly in 138 zones. This implies that (a) one cab makes approximately 2 pickups per hour, thus one unit of error is equivalent to half an hour of cab lost; (b) 10% market share (about 246 cab-hours per day); (c) only some fraction of gain could be utilized; (d) a cab-hour is valuable to be recovered; and (e) operating the system costs USD 3,000 per month.

Table 10. Scenario estimate of monthly net value for an operator with 10% market share (assumptions, not measured savings)

Scenario	Gain acted on	Net value per cab-hour	Monthly benefit	Monthly cost	Monthly net value
Worst case	10%	USD 15	≈ USD 11,100	USD 3,000	≈ USD 8,100

Scenario	Gain acted on	Net value per cab-hour	Monthly benefit	Monthly cost	Monthly net value
Base case	25%	USD 20	≈ USD 36,900	USD 3,000	≈ USD 33,900
Best case	40%	USD 25	≈ USD 73,900	USD 3,000	≈ USD 70,900

Sensitivity: The value is highly sensitive to what portion of the gain could be used by the operator, but even the worst-case scenario yields at least three times profit. The value will diminish if (i) summer accuracy turns out to be lower (untested), (ii) the demand shifts towards ride-hailing services; or (iii) there are many snow days. A pilot of tracking cruising should be conducted to substitute these assumptions. Priority: the impact and feasibility of week-ahead scheduling are the highest; the second place is given to rule-based staging; weather alert is relatively inexpensive but is supported by less robust evidence for rainy days. The complete roadmap is given in Section 9.

7. Interactive Application, Figure Index and Reproducibility

7.1 Interactive Dashboard

Dashboard and access: The interactive dashboard is a Streamlit application in the dashboard/ subdirectory of the codebase. It operates offline and locally once it has been built with a single data preparation step. It is completely based on the prepared dataset and project outputs. Six chapters of stories are told through the dashboard (data gathering and cleaning, EDA, predictions, forecasts, demand rules and business impact), with tabs, areas and zones selection and zone explorer that update the charts. Three live calculators compute on-the-fly the results. Demand predictor computes the results with the saved HistGB model for any zone and hour with weather and recent demand What-If sliders. Forecast planner uses the team tsa/baselines.py functions to forecast the demands of any area or zone with 80% range, the last-week baseline, actuals and a downloadable shift table for 24 hours and 7 days in advance starting at any time. Dispatch advisor matches the situation by (zone, day type, hour, temperature, rain, snow, holiday) and returns to the expected demand level, confidence, lift, a recommendation and a 24-hour forecast.

Architecture and dataflow: to produce outputs, build_data.py creates small files inside the directory dashboard/data/; to process user requests, app.py provides navigation; the directory views/ contains one file for each page. The analysis of data and graphs is done beforehand while predictions, forecasts, and rule matching are performed in real-time. Main and filtered states are presented in Figure 7. How to start the program: python3 -m venv .venv; source .venv/bin/activate; pip install -r dashboard/requirements.txt; python dashboard/build_data.py; cd dashboard; streamlit run app.py (http://localhost:8501).

Testing and limitations: Every page in Streamlit loaded properly (currently on version 1.67), and the selectors did their job, for instance the Airports 24-hour forecast indicated 6,539 pickups with a mean absolute error of 36.9. Changing the Dispatch Advisor's zone and day type had their effect in terms of matched rules and prognosis. The app was not deployed for public use, it uses scikit-learn 1.9 to upload the stored model, it

does not have a live data input (forecasts last for the period ending 1st of March 2026), the app works on one computer and not on host servers. No critical bugs have been discovered.

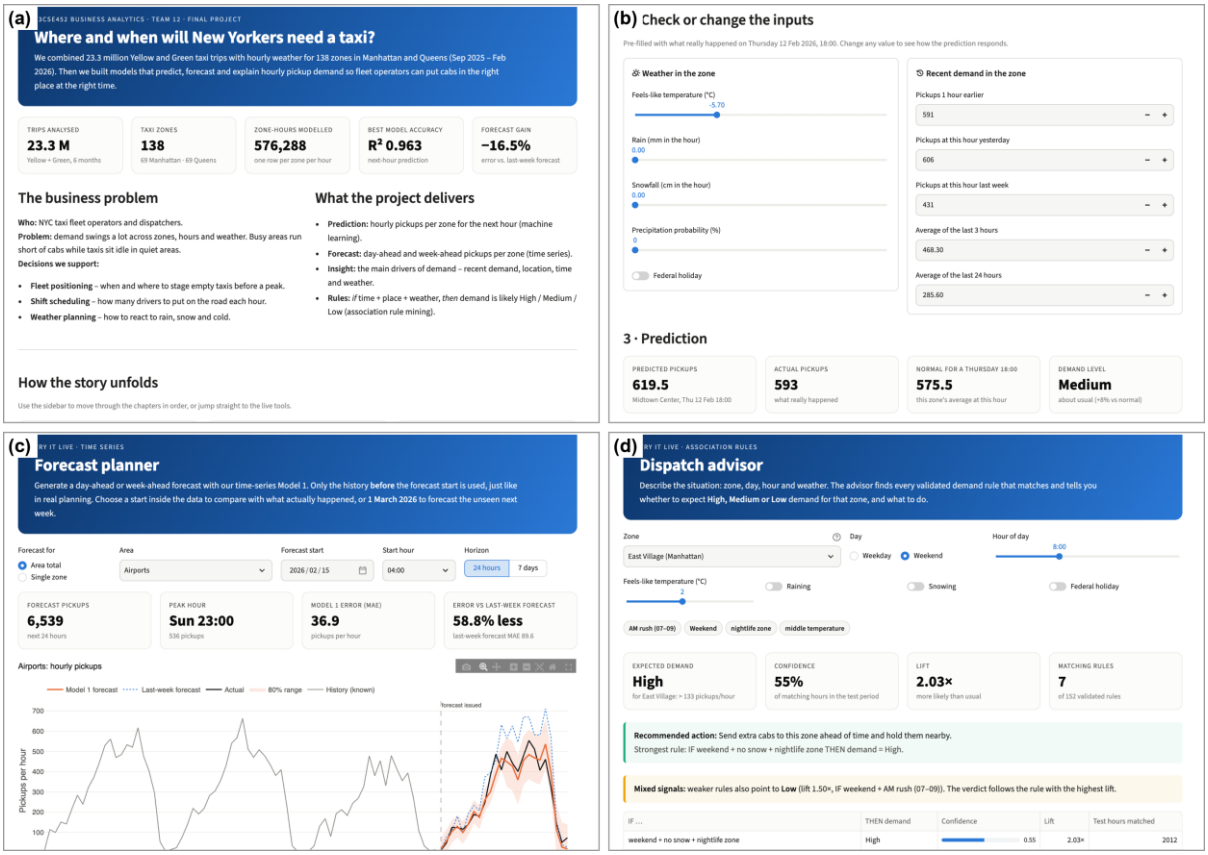


Figure 7. Streamlit dashboard: (a) home page (main state); (b) Demand predictor for Midtown Center, 12 Feb 2026 18:00, with a live prediction of 619.5 pickups against 593 actual; (c) Forecast planner filtered to Airports with a 24-hour horizon; (d) Dispatch advisor filtered to East Village on a weekend. Tool: Streamlit/Plotly.

7.2 Figure index and reproduction map

Table 11. Figure index and reproduction map

Figure	Title / tool	Section / key purpose	Source notebook / file
Fig. 1	Project workflow – Python/Matplotlib	1.1 – how units connect	Drawn by the team
Fig. 2	Pareto and zone clusters – Python/Matplotlib	3 – 16 zones = 50%; 3 zone types	scripts/EDA_visuallization.ipynb (C2, C8)
Fig. 3	Rain and snow ratios – Python/Matplotlib	3 – snow −75%, rain +14%	scripts/EDA_visuallization.ipynb (F5, F6)
Fig. 4	Importance and error by hour – Python/Matplotlib	4 – last hour dominates; evening errors	models/modeling/ (importance, error by hour)
Fig. 5	Day-ahead forecast week – Python/Matplotlib	5.1 – forecast tracks peaks	model_1/tables/ (bottom-up hourly intervals)
Fig. 6	Mining runtime and rule filtering – Python/Matplotlib	5.2 – 897 → 632 → 152 rules	data/processed/arm_findings/
Fig. 7	Dashboard states – Streamlit	7.1 – main and filtered states	dashboard/views/ (home, tool_*)

Figure	Title / tool	Section / key purpose	Source notebook / file
Figs. 8–13	Supplementary EDA, model and forecast plots	Appendix B	eda_arun.py; EDA_visuallization.ipynb; model_comparison_error_analysis.ipynb; model_1/figures
Figs. 14–16	Full-size dashboard screenshots	Appendix C	dashboard/views/ (ch2_eda, ch3_models, tool_predictor)
Fig. 17	Project board	Appendix D	GitHub Projects

7.3 Code and environment

Repository: https://github.com/BA-CapstoneProject-12/nyc-taxi-demand-forecasting.

The README provides the project structure, configuration and execution sequence. The key folders are data/raw, data/processed, data/dataset-documents; scripts (collection, processing, EDA, time-series and rule notebooks); models (Unit 2 and time_series/model_1); timeSeries (harmonic regression, MSTL + ETS); EDA_doc; ARM_doc; dashboard; and tests.

Software environment: Python ≥ 3.10; requirements.txt, requirements-time-series.txt and dashboard/requirements.txt; seed 42. Steps to reconstruct: data_collection.ipynb → data_processing_engineering.ipynb → EDA scripts → modelling_preparation_baseline.ipynb and three model notebooks → time_series_data_preparation.ipynb and run_time_series.py → association_rule_data_preparation.py and association_rule_mining.ipynb → build_data.py and streamlit run app.py.

License and access rights: TLC data belongs to NYC open data and Open-Meteo data has license CC BY 4.0; raw file of 3.2 GB is privately shared with team and can be replicated with the collection notebook; there is no personally identifiable information.

8. End-to-End Validation, Traceability and Contributions

8.1 Completion and validity

The full stated scope is achieved and verifiable from the repository: data collection/cleaning (cleaning_report.md), EDA (EDA_doc, 15 successful data checks), three models with saved predictions (models/modeling), unit 3 methods with executed notebook and outputs, and the Streamlit app. Validity threats: internal - fixed settings instead of tuning and the time series test interval has been seen beforehand; external - results apply to two boroughs, taxis only and winter time; for construct - pickups underestimate demand; for reproducibility - TLC can re-publish files, therefore file sizes, input hashes and versions of the packages are saved.

Objective / KPI	Analysis / method	Result / metric	Figure / artifact	Recommendation
Obj 1: Spatial concentration and priority zones	Sec 3 – EDA	Manhattan accounts for 90.6% of pickups (19.7M trips); 55	Figs. 2–3; EDA analysis	R1

Objective / KPI	Analysis / method	Result / metric	Figure / artifact	Recommendation
		Yellow zones account for 86.3% of total volume.		
Obj 2: Airport demand patterns	Sec 3 – EDA	LGA peak: 258.6 trips/hr at 14:00; JFK peak: 380.9 trips/hr at 16:00. Two airport zones account for 6.5% of pickups	Figs. 2–3; EDA analysis	R2
Obj 3: Weather-related demand variation	Sec 3 – EDA	Morning rain increases demand by 14.6%–22.1%; heavy snow reduces demand by 40%–70%.	EDA weather analysis	R3
Obj 4: Multi-day demand forecasting	Sec 5.1 – Time-Series Analysis	Strong 24-hour and 168-hour seasonal cycles support daily and weekly demand forecasting.	Fig. 5; Model 1 results	R4
Obj 5: Demand pattern discovery	Sec 5.2 – FP-Growth / ARM	Weekday morning commute rules: 70.3%–73.3% confidence and 2.60–2.71 lift. Weekend nightlife rule: 54.1% confidence and 2.38 lift.	Fig. 6; ARM findings	R5, R6

Table 12. Objective-to-evidence traceability matrix

8.2 Project tracking and individual work

Task assignment was managed using the GitHub Projects board at: https://github.com/orgs/BA-CapstoneProject-12/projects/3. Issues were created on EDA issues #59 to #66 and rules issues #30 to #33 in respective branches and merged via pull requests #57, #58, #67, #68, #70, #73, #74; see Appendix D.

Table 13. Team members, assigned work and evidence

Name/ Register No.	Assigned work	Evidence link / artifact
Adithya Ajay, CB.SC.U4CSE23102	Data collection; time-series data preparation; Time-Series Model 1	data_collection.ipynb (PR #57, 30 Sep 2026); time_series_data_preparation.ipynb (9–10 Oct); model_1/ (PR #74, 11 Oct)
Amrith B, CB.SC.U4CSE23105	EDA I (zones, time, weather, features); interactive Streamlit dashboard	EDA commits #59–#62 (3 Oct); dashboard/ (commit d8fbe8f, 7 Nov)
Arun N, CB.SC.U4CSE23406	EDA II (data checks, calendar, holidays, hotspots); rule-mining data preparation	eda_arun.py, issues #63–#66 (PR #68, 3 Oct); association_rule_data_preparation.py, issue #30 (10 Oct)
Sadhana T P, CB.SC.U4CSE23441	Predictive modelling and evaluation; harmonic regression and MSTL + ETS models	models/ notebooks (4 Oct); timeSeries/ notebooks and results (11 Oct)
Pranathi Nibhanipudi, CB.SC.U4CSE23643	Cleaning, preprocessing, feature engineering; association rule mining	data_processing_engineering.ipynb (PR #58, 30 Sep); association_rule_mining.ipynb, ARM_doc (PRs #70, #73, 11 Oct)

8.3 AI use and originality declarations

Use of Generative AI: Generative AI tools like Antigravity, Claude, and Codex were used to assist in code writing and documentation. A generative AI writing assistant (Claude) was used to arrange, write and prepare this report using the repositories, findings and presentations established by the team. The report was confirmed through validation against project results. The complete report was proofread by all the team members to verify the correctness.

Originality of the contents: The results of analysis, software and application developed including dashboards were created by the team and the content from already existing academic sources was referenced.

9. Conclusion, Recommendations and Roadmap

It is evident from the data analysis presented above in Table 12 that the five objectives were fully accomplished. To summarize, the self-compiled dataset of 23.3 million trips was obtained, the EDA was conducted in order to understand the time, place and the impact of weather on demand, the next hour predictions reached 37.9% improvement, one-day ahead predictions were 16% better than baseline, discovery of 152 demand rules was accomplished, and the interactive dashboard was created.

#	Recommendation	Expected KPI effect	Stakeholder	Feasibility
R1	Concentrate 85%–90% of active vehicles in core Manhattan / high-volume Yellow zones; reduce unnecessary cruising in 42 near-zero-demand outer zones.	Lower unbilled fuel consumption and idle mileage by focusing on high-volume corridors.	Operations manager, dispatch team	High
R2	Stage airport vehicles 30–60 minutes before the LGA 14:00 and JFK 16:00 demand peaks.	Capture more high-fare airport trips and reduce passenger waiting time.	Airport dispatch team, drivers	High
R3	Deploy additional drivers during morning rain; reduce the active fleet by approximately 50% during heavy-snow warnings.	Capture rain-related demand increases while reducing idle costs and weather-related operational risks.	Operations manager	Medium
R4	Plan driver shifts and fleet maintenance 1–7 days ahead using the identified 24-hour and 168-hour seasonal demand cycles.	Improve multi-day vehicle availability and schedule maintenance during lower-demand periods.	Operations manager	High
R5	Pre-position vehicles in residential and commuter zones before 07:00 on weekdays.	Target demand periods with 70.3%–73.3% confidence of High demand and 2.60–2.71 lift.	Dispatch team	High
R6	Redeploy evening cabs to nightlife clusters such as Lower Manhattan and Williamsburg from 23:00–04:00 on Fridays and Saturdays.	Target weekend nightlife periods with 54.1% confidence of High demand and 2.38 lift.	Dispatch team, drivers	High

Table 14. Prioritised recommendations

Roadmap:

Immediate (0-1 month) - Launch the Streamlit application on internal servers; train drivers; perform R1-R3 (one analyst and one lead dispatcher) so that weekly prediction error remains at test level.

Mid-term (1-6 months) - Make automated systems for TLC and climate forecasts monthly, putting in place overrides for the snow alerts (R4), doing the test case (R5), and determining the cruising before and after.

6-12 Months - Retrain the model using data for one year including all five boroughs and update the rules per season.

Future improvements can include using multi-year data along with the event calendar, considering snow and holidays in forecasting, incorporating LSTM, and incorporating ride-hailing data and drop-off points.

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Appendix A: Extended Data Dictionary and Validation Records

Collected trip file (23,318,979 rows × 14): taxi_type, pickup_datetime, dropoff_datetime, PULocationID, DOLocationID, pickup_borough, pickup_zone, pickup_zone_latitude, pickup_zone_longitude, pickup_hour, apparent_temperature (°C), rain (mm), snowfall (cm), precipitation_probability (%). Supporting files: taxi_zones_manhattan_queens.csv (138 × 6), weather_hourly_by_zone.csv (599,472 × 9, with the grid cell used), taxi_collection_summary.csv (12 × 14 audit of each monthly file).

Table 15. Monthly trips collected by source file

Month	Yellow rows in file	Yellow kept	Green rows in file	Green kept
2025-09	4,251,015	4,048,166	48,893	40,059
2025-10	4,428,699	4,234,338	49,416	40,573
2025-11	4,181,444	3,982,593	46,912	38,373
2025-12	4,305,006	4,096,636	48,236	39,806
2026-01	3,724,889	3,525,433	40,272	32,792
2026-02	3,399,866	3,209,710	37,373	30,500
Total	24,290,919	23,096,876	271,102	222,103

Table 16. Federal holiday effects on daily pickups against matched normal weekdays

Date	Holiday	Actual trips	Normal mean	Change
2025-10-13	Columbus Day	104,735	114,190	−8.3%
2025-11-11	Veterans Day	123,875	130,882	−5.4%
2025-11-27	Thanksgiving Day	86,029	155,811	−44.8%
2025-12-25	Christmas Day	60,817	143,269	−57.6%
2026-01-01	New Year's Day	102,562	143,269	−28.4%
2026-01-19	Martin Luther King Jr. Day	93,566	93,566	0.0%
2026-02-16	Washington's Birthday	81,311	90,738	−10.4%

No synthetic data, survey, or consent process was used. Input hashes and package versions for the rule stage are in data/processed/association_rules/arm_input_manifest.json and arm_environment.json.

Appendix B: Additional Figures and Supplementary Analysis

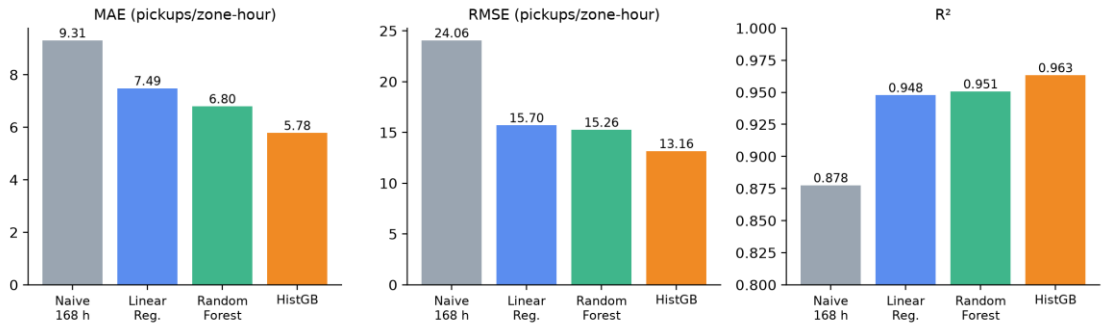


Figure 8. Test MAE, RMSE and R² of the baseline and three models. Source: models/modeling/model_comparison_final.csv.

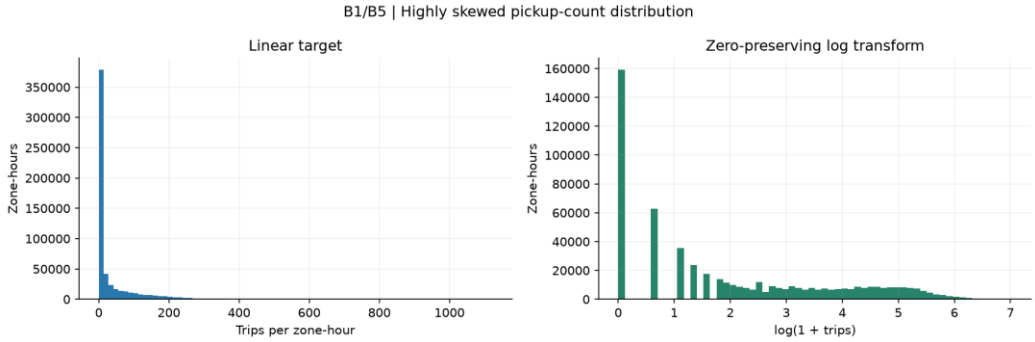


Figure 9. Distribution of pickups per zone-hour, raw (left) and log(1 + x) (right). Source: scripts/eda_arun.py.

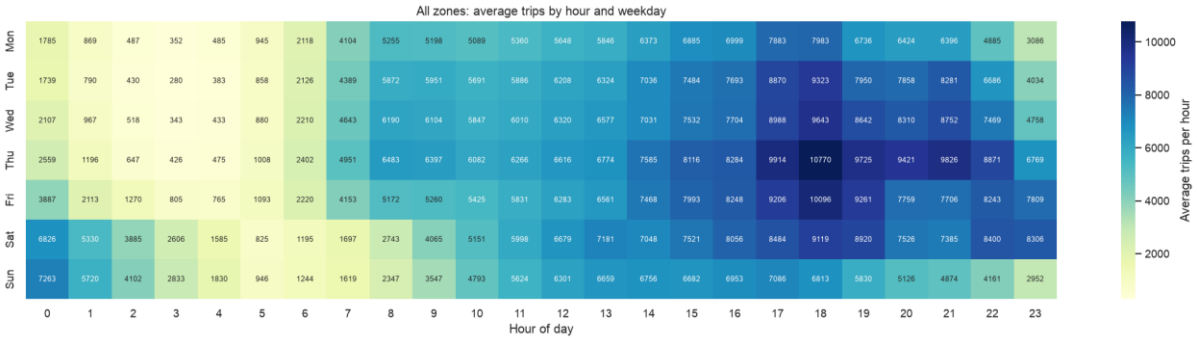


Figure 10. Average pickups per hour by weekday and hour of day (holidays excluded). Source: scripts/EDA_visuallization.ipynb.

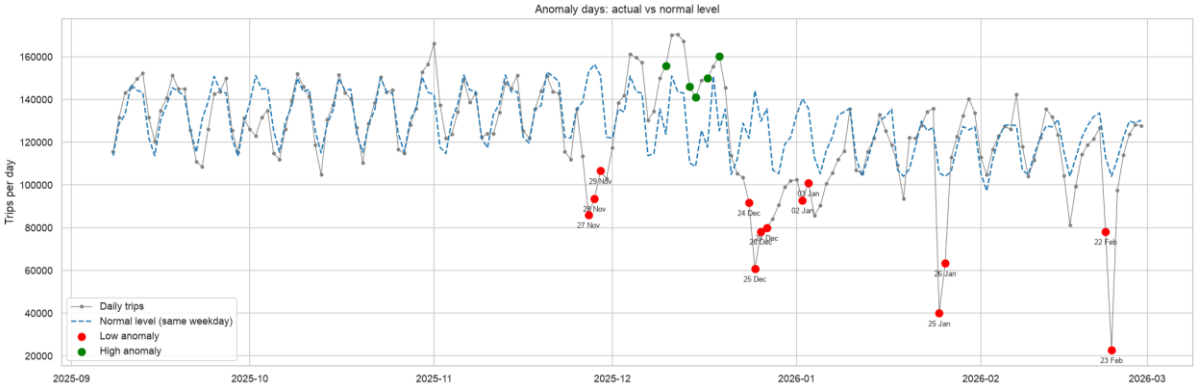


Figure 11. Daily trips against the normal level for the same weekday; red = unusually low, green = unusually high. Source: scripts/EDA_visuallization.ipynb.

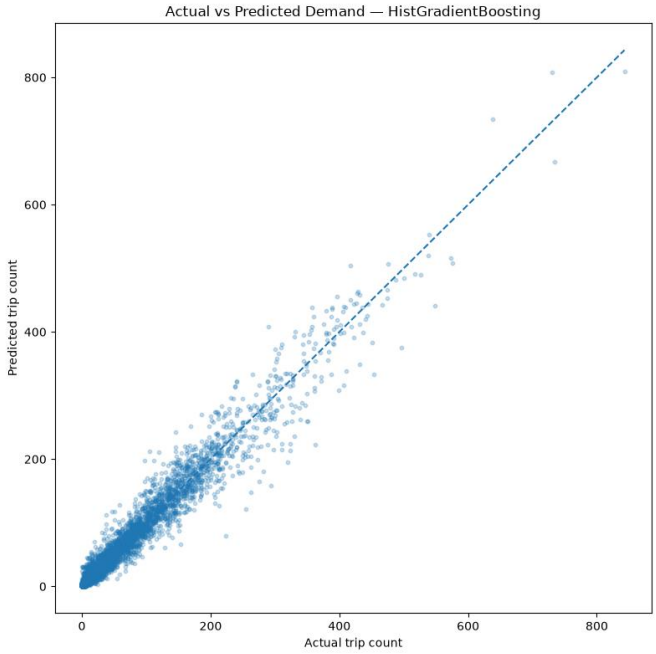


Figure 12. Actual against predicted pickups per zone-hour, HistGradientBoosting, test period. Source: models/model_comparison_error_analysis.ipynb.

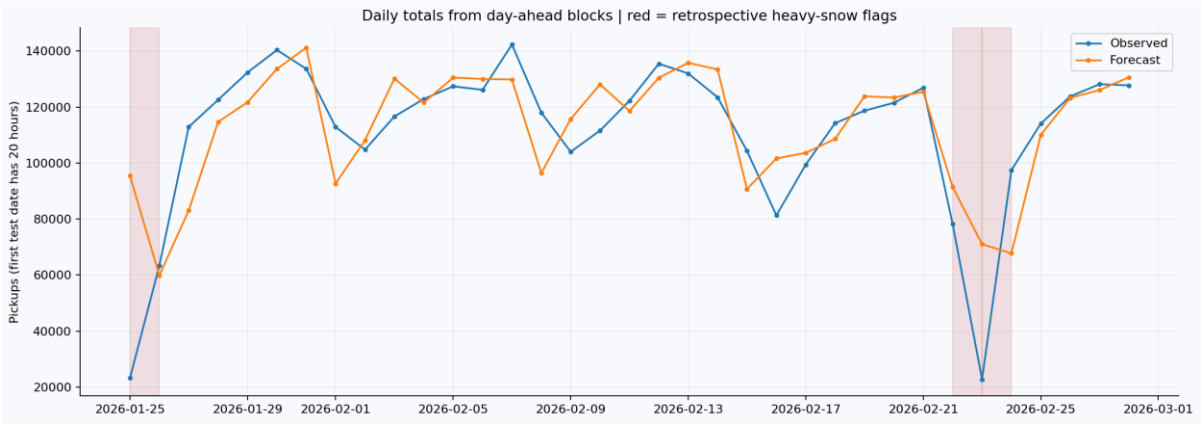


Figure 13. Daily totals from day-ahead forecasts against actuals; red bands mark heavy-snow days. Source: models/time_series/model_1/figures/11_event_stress_test.png.

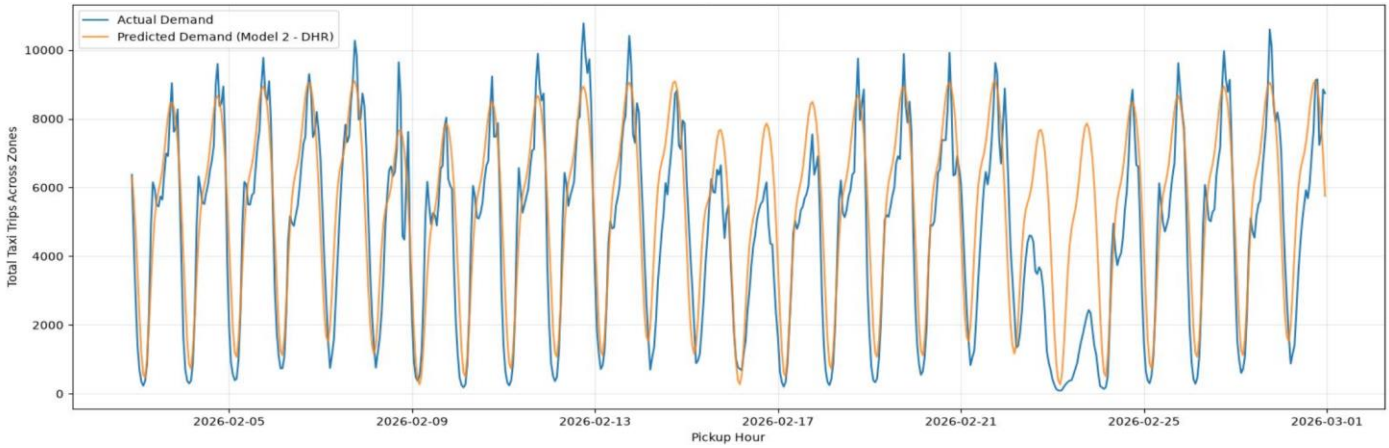


Figure 14. DHR Actual vs Predicted Demand

Table 17. Weather ablation and train/test scores

Model / variant	Train MAE	Test MAE	Train R^2	Test R^2
Linear Regression	7.31	7.49	0.955	0.948
Random Forest	6.20	6.80	0.969	0.951
HistGB with weather	5.54	5.78	0.972	0.963
HistGB without weather	5.54	5.67	0.972	0.965

Table 18. Day-ahead zone forecast error by zone group (Model 1 vs weekly naive)

Zone group	Weekly naive MAE	Model 1 MAE	R^2
High-volume zones	25.84	21.78	0.856
Medium-volume zones	7.50	6.13	0.800
Low-volume zones	2.18	1.83	0.522
Sparse zones	0.57	0.53	0.140
Intermittent pattern	0.66	0.59	0.211

Appendix C: Implementation Details and Dashboard User Guide

Installation and execution of commands as follows: pip install -r requirements.txt (requirements-time-series.txt as well for Unit 3). Check commands: python models/time_series/model_1/test_time_series.py; python scripts/association_rule_data_preparation.py --validate-only; python -m unittest discover -s tests -v. Instruction for Streamlit: visit http://localhost:8501 using sidebar. Demand predictor allows selection of zone, date and time, while manipulating the sliders for "what if" scenarios. Forecast planner can be utilized by selecting area/zone, starting time, duration and making comparisons with the baseline before saving the final result. Dispatch advisor assists in selecting zone, day type, timing, and weather before seeing correspondent

rules and forecasts for the following 24 hours. The commands for Windows environment are in dashboard/README.md.

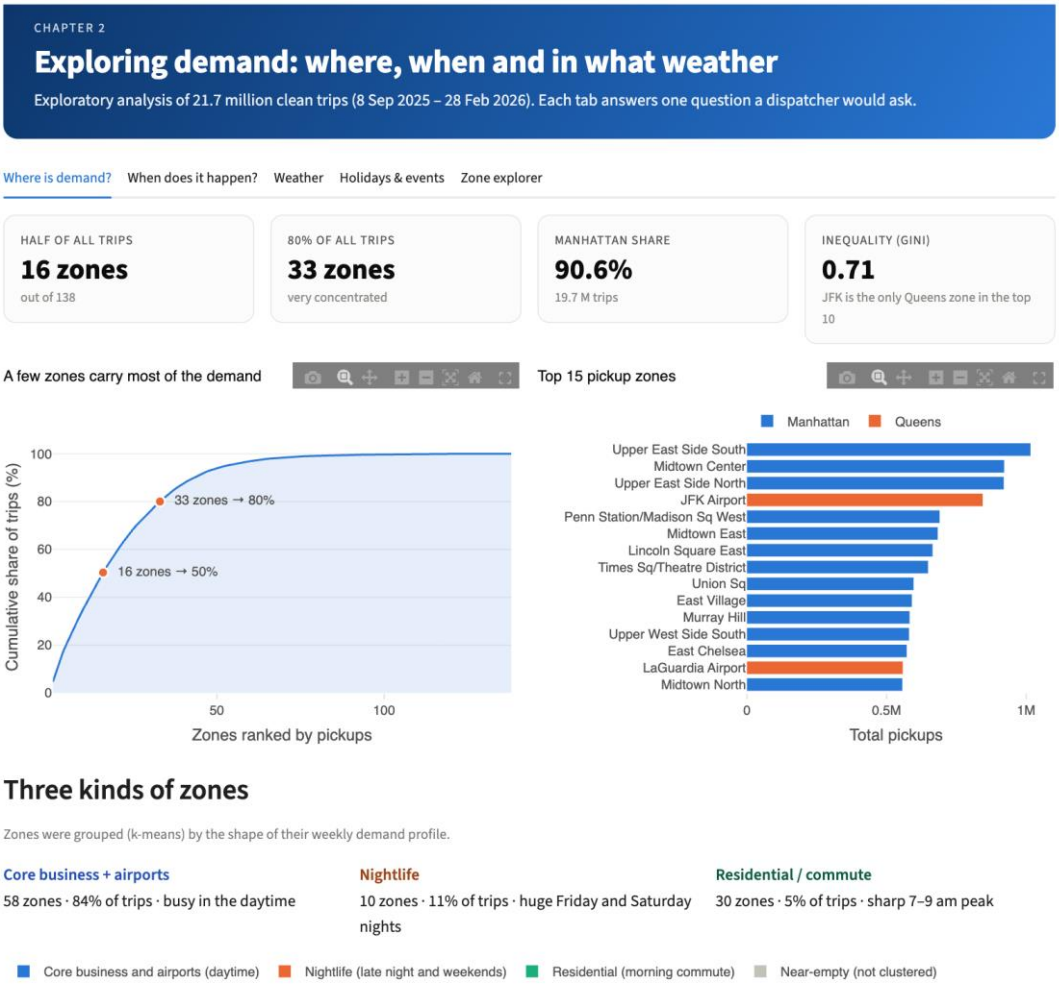


Figure 14. Streamlit chapter 2, Exploring demand: KPIs, Pareto curve, top zones and zone-cluster map, with tabs for time, weather, holidays and a zone explorer.

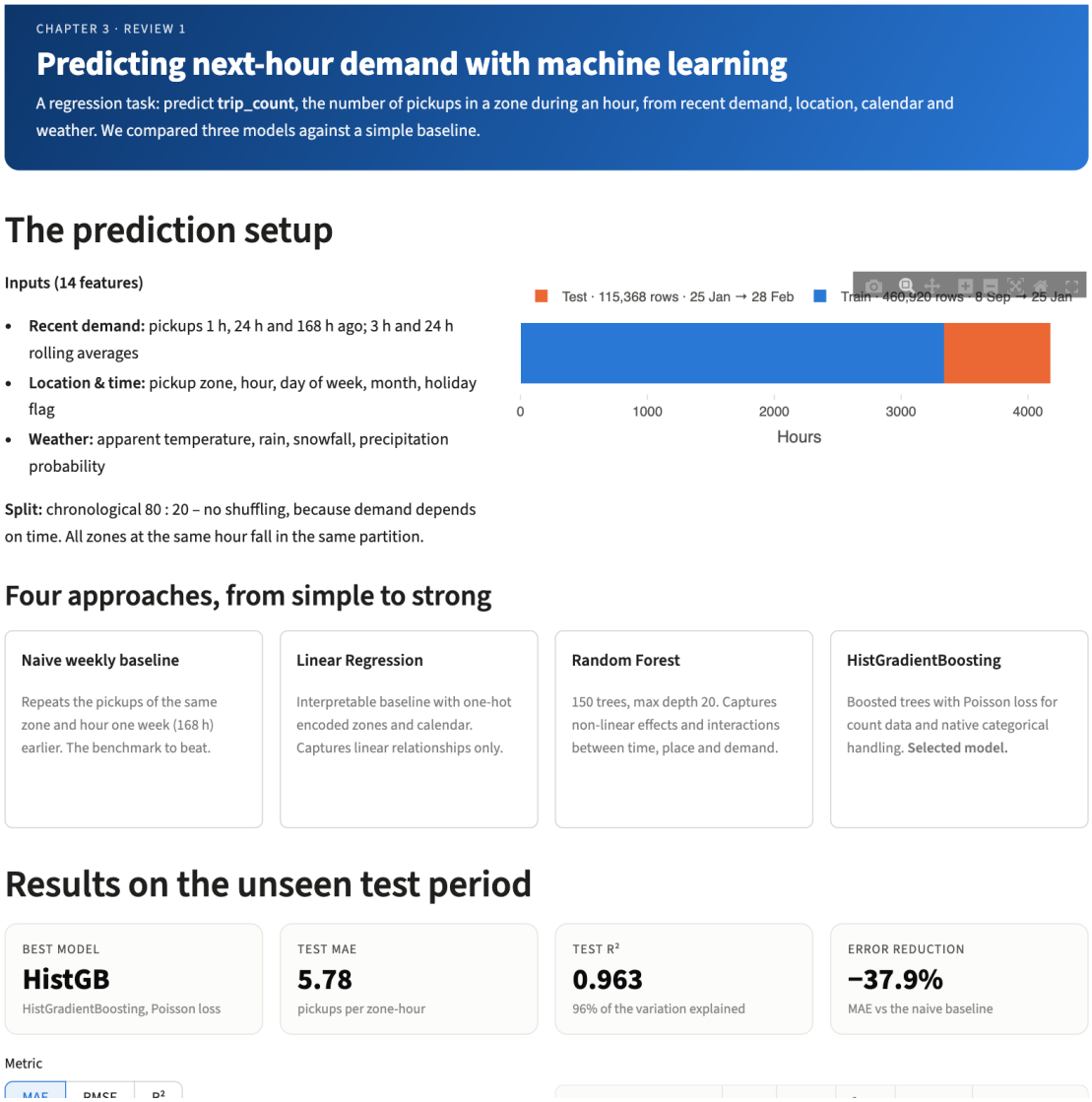


Figure 15. Streamlit chapter 3, Predicting demand: setup, models, test results with a metric selector, and overfitting check.

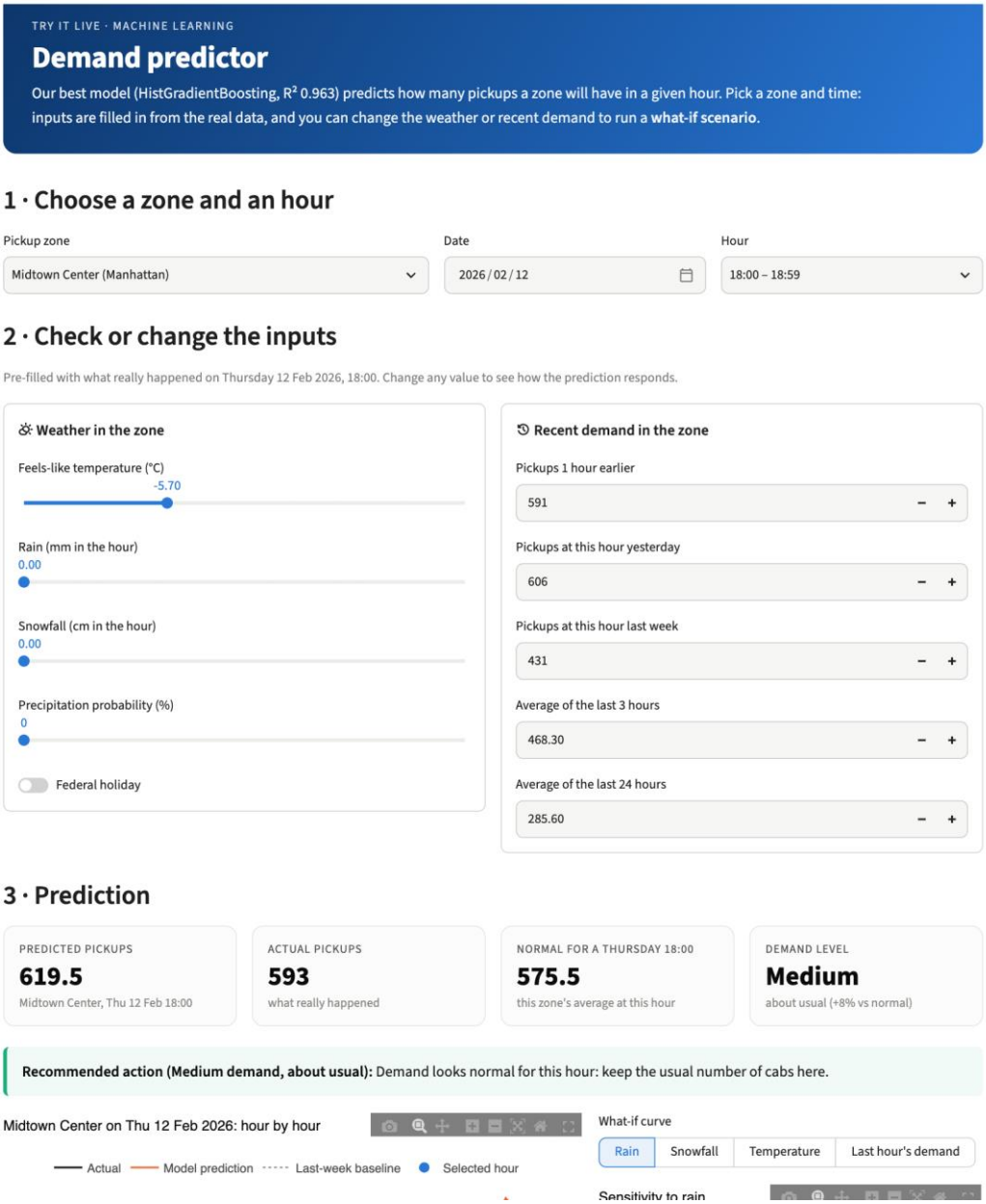


Figure 16. Streamlit Demand predictor for Midtown Center, 12 Feb 2026 18:00: live prediction 619.5 against 593 actual.

Appendix D: Tracking-Board Evidence, Contributions and Declarations

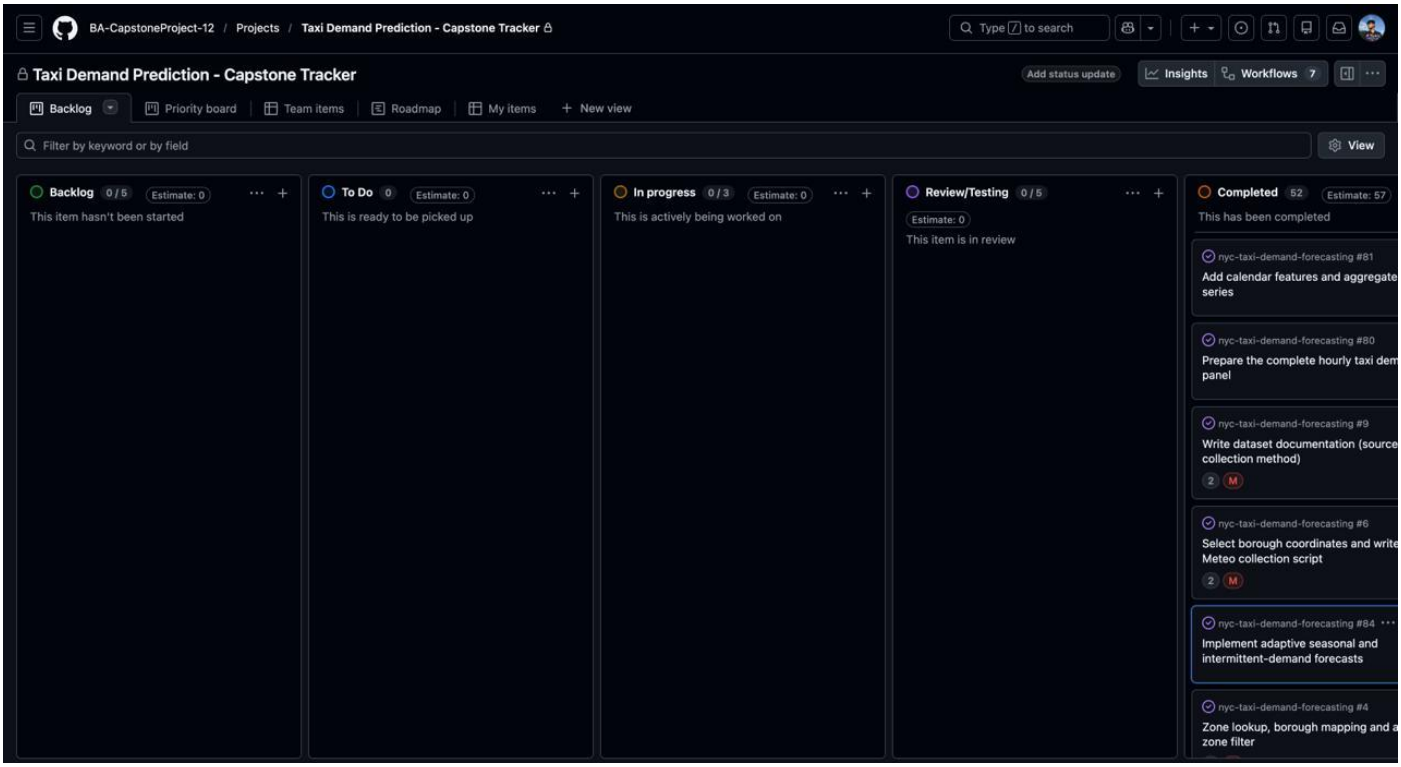


Figure 17. Team GitHub Projects board showing tasks by stage. Source: https://github.com/orgs/BA-CapstoneProject-12/projects/3.

Contribution records are in Table 13, with full commit, branch and pull-request history in the repository. The signed declaration is in the front matter and the AI-use disclosure in Section 8.3.